Artifact Requirements

This file summarizes the requirements for the artifact accompanying "Suborbital and Orbital Real-Time Localization of Gamma-Ray Bursts via Utility-Guided Coordination." The complete installation and reproduction instructions are in `README.md`.

Evaluation paths

The artifact provides three evaluation paths:

1. Reproduce Figures 4–11 from the paper results (README Section 2).
2. Regenerate the training and evaluation results (README Section 3).
3. Optionally rerun the platform-sensitive mapping benchmark for Figure 4 (README Section 4).

Figures 1–3 are explanatory diagrams and are not generated by the artifact.

Hardware

- Operating system: Linux.
- Architecture for the complete Sections 1–3 workflow: ARM64 (`aarch64`).
- Reference ARM platform: NVIDIA Jetson Orin NX with eight ARM Cortex-A78AE v8.2 CPU cores, CPU frequency fixed at 1.5 GHz, and 16 GB DRAM. The GPU is not used.
- Architecture for the optional Intel benchmark: Intel/AMD x86-64.
- Reference Intel platform for Figure 4: Intel Xeon Gold 5118 at 2.3 GHz with 256 GB DRAM.
- Storage: at least 65 GB after extraction, plus temporary space for the 16.2 GB downloaded archive.
- Non-commodity peripherals: none.

The timing results are platform-sensitive. Exact timing comparisons require the corresponding reference platform. Functional execution on another compatible platform does not imply matching execution times.

Software

- Git.
- Bash and standard Linux command-line tools.
- `wget` or `curl`, `tar`, and `md5sum`.
- Conda/Miniconda.
- Python and Python packages specified by:

    - `cosipy-312-deps.yaml` for ARM64; and
    - `cosipy-312-intel.yml` for the optional Intel benchmark.
- Jupyter Notebook for result visualization.
- CMake and Make.
- A C++17 compiler with OpenMP support.
- LEMON Graph Library 1.3.1.
- HDF5 with the C++ and high-level libraries enabled.
- HighFive headers.
- The Bitshuffle HDF5 plugin supplied by the Python `hdf5plugin` package.
- The `tsmap-artifact` branch of COSIpy, installed as described in `README.md`.

The two Conda YAML files are the authoritative package specifications for their respective Python environments.

Data and network access

- The source repository is approximately 200 MB.
- The separate `transients.tar.gz` archive is approximately 16.2 GB.
- The extracted detector models and transient datasets require approximately 25–30 GB.
- Full experiment runs can generate approximately 30 GB of maps.
- Network access is required during installation to obtain the repositories, dependencies, and Zenodo data archive. The experiments themselves can run offline after these resources have been installed.

Expected skills

Reviewers should be comfortable with a Linux shell, Conda environments, CMake-based C++ builds, environment variables, and Jupyter notebooks. No specialized astronomy knowledge is required to execute the artifact.

Approximate execution time

The times below are the estimates given in `README.md` for the reference platforms:

- Reproducing figures from distributed results: minutes.
- Full training phase: approximately 18 hours.
- Full evaluation phase: approximately 21 hours.
- Optional Intel mapping benchmark: approximately 20 minutes.
- Optional Jetson mapping benchmark: approximately 10 minutes.

Actual times depend on the platform, storage system, and available CPU resources.